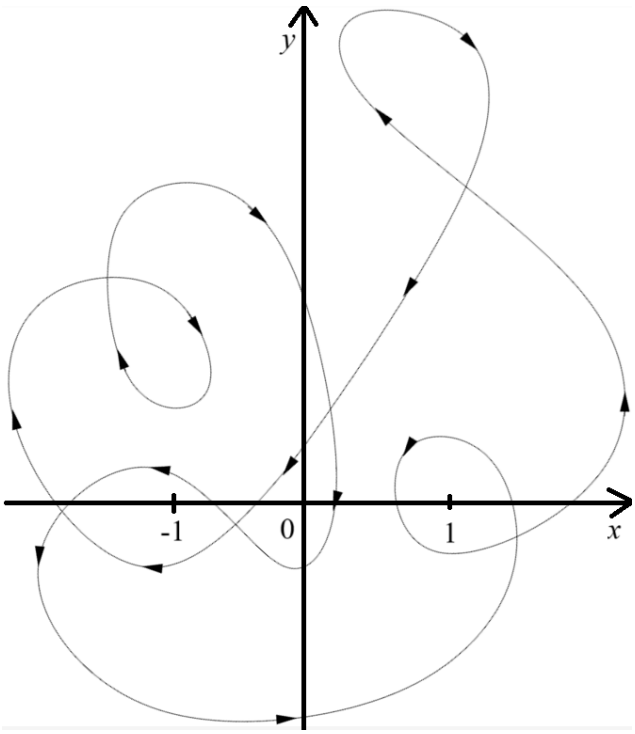
Vector Calculus and Complex Analysis Classwork (2025/5/12)

Name:

Student ID:

Let C be the following curve. Calculate $\int_C \frac{e^z}{z^3 - z^2 - z + 1} dz$.



Solution

$$\int_C \frac{e^z}{z^3 - z^2 - z + 1} dz = 2\int_{|z-1|=1} \frac{e^z}{(z-1)^2(z+1)} dz = 4\pi i \frac{d}{dz} \frac{e^z}{(z+1)}\bigg|_{z=1} = 4\pi i \frac{(z+1)e^z - e^z}{(z+1)^2}\bigg|_{z=1} = e\pi i$$